

AE 696 Homework 1

Markov Property and MDP Formulation

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Total: 100 points

Instructions. Submit one PDF. No code is required. Use the coordinate and reward conventions stated below; do not introduce an alternative convention. Show the requested equations and calculations. Round final numerical answers to four decimal places.

Learning goals. Markov state assessment, complete MDP specification, stochastic transitions, discounted return, and information-state construction.

Problem 1: Markov state assessment

[20 points]

For each system, report: (i) **Yes or No**, whether the proposed state is Markov when conditioned on the current action; (ii) the **minimum state** using only the variables shown; and (iii) the **equation term** that determines your answer. Each system is worth 5 points.

1.1 A thermostat obeys

yes, Markov

$$T_{t+1} = 0.9T_t + 0.1T_{\text{out}} + 2a_t + w_t,$$

where T_{out} is a known constant, $a_t \in \{0, 1\}$ is the current action, and w_t is independent noise.

Proposed state: $S_t = T_t$. *Term = $0.9T_t$. $1T_{\text{out}}$ = known, $2a_t$ = curr. action, w_t = independent*

*constant
current action independent / new sample
not needed*

min state: $S_t = T_t$

1.2 A point-mass robot obeys

not Markov

$$x_{t+1} = x_t + \Delta t v_t, \quad v_{t+1} = v_t + \Delta t a_t.$$

Proposed state: $S_t = x_t$.*determining terms:* *$\Delta t v_t$ in eqn 1, & v_t in eqn 2**if same pos but diff velocity the next state would be diff**min state: $S_t = \begin{bmatrix} x_t \\ v_t \end{bmatrix}$* 1.3 A grid robot has proposed state $S_t = c_t$, its current cell. The map is fixed and*yes, Markov*

$$\Pr(c_{t+1} = c' \mid c_t, a_t) = P(c' \mid c_t, a_t),$$

where the same transition table P is used at every time step.

determining term: $\Pr(c_{t+1} = c' \mid c_t, a_t) = P(c' \mid c_t, a_t)$ *min state: $S_t = c_t$* *the curr. cell determines the action, no history needed*1.4 A battery model obeys w_t is needed for h_{t+1} & R_{t+1} - *not Markov*

$$z_{t+1} = z_t - c|u_t|, \quad h_{t+1} = h_t - \epsilon|u_t|, \quad R_{t+1} = -(1 - h_t),$$

where z_t is state of charge and h_t is state of health. Proposed state: $S_t = z_t$. *determining terms: $h_{t+1} = h_t - \epsilon|u_t|$, $R_{t+1} = -(1 - h_t)$*

min state $S_t = \begin{bmatrix} z_t \\ h_t \end{bmatrix}$

Problem 2: Stochastic grid navigation MDP

[35 points]

Rows are numbered top to bottom and columns left to right. The start is $S = (1, 1)$, the terminal goal is $G = (3, 4)$, and obstacles are $(1, 4)$ and $(2, 2)$. Obstacle cells are not states.

1	S			X
2		X		
3				G
4				
	1	2	3	4
	column			

At each nonterminal state, $\mathcal{A} = \{U, D, L, R\}$. The intended direction occurs with probability 0.8 and each perpendicular direction with probability 0.1; there is no backward slip. A sampled direction that hits a wall or obstacle leaves the robot in the same cell.

The realized reward is +10 for entering G , -10 when the sampled direction hits a wall or obstacle, and -1 for any other nonterminal move. The episode ends immediately upon entering G , and $\gamma = 0.95$.

Let $F(s, d)$ be the neighboring cell in direction d if that cell is valid, and let $F(s, d) = s$ otherwise.

$$\mathcal{S} = \{(r, c) \mid r \in \{1, 2, 3, 4\}, c \in \{1, 2, 3, 4\}, (r, c) \notin \{(1, 1), (2, 2)\}\}$$

2.1 Write $\mathcal{S}$ in set notation and give $|\mathcal{S}|$. $|\mathcal{S}| = (\#r) \cdot (\#c) - (\# \notin) = 4 \cdot 4 - 2$ $|\mathcal{S}| = 14$ [6 points]

2.2 Write $\mathcal{A}$. $\mathcal{A} = \{U, D, L, R\}$ [4 points]

2.3 Using F , indicator functions, and the three direction probabilities, write one formula for $P(s' \mid s, a)$. [8 points] $P(s' \mid s, a) = .8 \cdot \mathbb{1}\{s' = F(s, a)\} + .1 \cdot \mathbb{1}\{s' = F(s, d_1, a)\} + .1 \cdot \mathbb{1}\{s' = F(s, d_2, a)\}$

2.4 For $s = (1, 1)$ and $a = R$, list every possible pair (S_{t+1}, R_{t+1}) and its probability. Verify that the probabilities sum to one. $P(s' \mid (1, 1), R) = .8 \cdot \mathbb{1}\{s' = (1, 2)\} + .1 \cdot \mathbb{1}\{s' = (1, 1)\} + .1 \cdot \mathbb{1}\{s' = (2, 1)\}$ [7 points]

$$\sum_{s'} (.8 \cdot \mathbb{1}\{s' = (1, 2)\} + .1 \cdot \mathbb{1}\{s' = (1, 1)\} + .1 \cdot \mathbb{1}\{s' = (2, 1)\}) = 1$$

2.5 Write the reward rule as a piecewise function and identify the terminal-state convention. [6 points] $R(s, a, s') = \begin{cases} +10 & s' = G \\ -10 & s' = s \text{ (blocked movement) or } F(s, a) = s \\ -1 & \text{otherwise} \end{cases}$

2.6 Is the current grid cell Markov under the stated assumptions? Circle **Yes** or **No**. Then circle the correct reason: (A) the next outcome depends on the full path, (B) the distribution of (S_{t+1}, R_{t+1}) is fixed by (S_t, A_t) , or (C) the reward is always deterministic. **no b/c there are possibilities 80%, 10%, 10%** [4 points]

Problem 3: Nominal routes and discounted return

[25 points]

This problem continues to use the **same grid map, coordinate convention, start state, goal state, obstacle cells, reward rule, and discount factor from Problem 2**. Override only the stochastic slippage: for Problem 3, assume every intended action succeeds. Starting from $(1, 1)$, use the exact action sequences

Route A: R, R, D, D, R,

Route B: D, D, R, D, R, R, U.

Use the original rewards and $\gamma = 0.95$.

3.1 Write the state trajectory for Route A. [5 points]

$$(s_0, s_1, s_2, s_3, s_4, s_5) = ((1, 1), (1, 2), (1, 3), (2, 3), (3, 3), (3, 4))$$

3.2 Write the state trajectory for Route B. [5 points]

$$(s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7) = ((1, 1), (2, 1), (3, 1), (3, 2), (4, 2), (4, 3), (4, 4), (3, 4))$$

3.3 Write the reward sequence for each route. [4 points]

$$(R_1, R_2, R_3, R_4, R_5) = (-1, -1, -1, -1, +10)$$

$$(R_1, R_2, R_3, R_4, R_5, R_6, R_7) = (-1, -1, -1, -1, -1, -1, +10)$$

3.4 Compute G_A and G_B . [8 points]

$$G_A = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots \quad G_A = 4.4352, \quad G_B = 2.0538$$

3.5 Circle the correct statement: (A) these sequences are complete feedback policies for the stochastic MDP, or (B) they are nominal open-loop plans and do not specify an action after an off-route slip. Then state which route has the larger nominal return and report $G_A - G_B$. [3 points]

no solution if there is a stochastic slip.

$$G_A - G_B = 2.3814$$

Problem 4: Quantitative reward and state questions

[20 points]

This problem continues to use the **same grid map, coordinate convention, start state, goal state, obstacle cells, and stochastic transition model from Problem 2**. Route A means the exact Route A defined in Problem 3. A task changes a reward value only when it states that change explicitly.

Each task is worth 5 points.

- 4.1 Keep Route A, $\gamma = 0.95$, and the +10 goal reward, but change every ordinary step reward from -1 to -0.1. Compute the new nominal return.

$$G_A = -1 - .1\gamma - .1\gamma^2 - .1\gamma^3 + 10\gamma^4 \rightarrow G_A = 7.7741$$

- 4.2 From (1,1), command R. Compute $\mathbb{E}[R_{t+1} \mid S_t = (1,1), A_t = R]$ first with collision reward -10, then with collision reward -2. All other rewards and transition probabilities remain unchanged.

$$\mathbb{E}(R_{t+1}) = \sum_i p_i R_i = .8(-1) + .1(-10) + .1(-1) = -1.9000, \quad .8(-1) + .1(-2) + .1(-1) = -1.1000$$

- 4.3 A robot obeys

$$x_{t+1} = x_t + v \cos \theta_t \Delta t, \quad y_{t+1} = y_t + v \sin \theta_t \Delta t, \quad \theta_{t+1} = \theta_t + \omega_t \Delta t.$$

Circle the Markov state: (A) (x_t, y_t) or (B) (x_t, y_t, θ_t) . Name the variable omitted by the non-Markov choice.

θ_t is omitted from A

- 4.4 The position sensor has a one-step delay. At time t , the controller observes $x_{t-1} = (1,1)$ and knows that $a_{t-1} = R$. Using the grid transition model, write the belief $b_t(s) = \Pr(X_t = s \mid x_{t-1}, a_{t-1})$ for every state with nonzero probability. Also write the two-component information state I_t used to compute this belief.

$$b_t(s) = \begin{cases} .8 & s = (1,2) \\ .1 & s = (2,1) \\ 0 & \text{otherwise} \end{cases} \quad \sum_{s \in S} b_t(s) = .8 + .1 + .1 = 1 \quad I_t = (x_{t-1}, a_{t-1}) = ((1,1), R)$$

Submission checklist and grading summary

Confirm that your PDF labels all answers, shows both return calculations, uses the stated coordinate and reward conventions, and reports numerical results to four decimal places.

Problem	Focus	Points
1	Markov state assessment	20
2	Complete MDP formulation	35
3	Trajectories and return	25
4	Quantitative reward and state questions	20
Total		100